

## GRANBY, QC, Canada

WMO: 710367

Lat: 45.3727N

Lon: 72.7743W

Elev: 282

StdP: 14.55

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.4      | -8.4 | -20.5                         | 1.8 | -12.8 | -15.9 | 2.3 | -7.3 | 20.1                  | 22.5 | 18.1 | 17.9 | 3.8                   | 220  | 0.476 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.8                   | 85.7            | 71.5 | 83.0 | 70.0 | 80.4 | 68.4 | 74.2                | 81.8 | 72.5 | 79.7 | 71.0 | 77.4 | 6.1                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 71.8                            | 118.8 | 78.2 | 70.2 | 112.2 | 76.4 | 68.7 | 106.5 | 75.0 | 37.9          | 82.2 | 36.4 | 79.8 | 35.0 | 77.0 | 81.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 15.9              | 13.5 | 11.6 | -18.1                      | 90.5 | 8.1                | 2.3 | -23.9                                              | 92.1 | -28.7      | 93.5 | -33.2      | 94.7 | -39.1      | 96.4 |  |  |
|                   |      |      | -19.0                      | 77.2 | 8.3                | 2.4 | -25.0                                              | 78.9 | -29.8      | 80.3 | -34.5      | 81.6 | -40.6      | 83.3 |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.3   | 16.1  | 18.2  | 27.8  | 42.1  | 56.2  | 64.0  | 69.8  | 67.4  | 60.0  | 48.4  | 35.4  | 24.3  |
|                                                                        | DBStd       | 20.99  | 14.66 | 11.98 | 11.80 | 8.41  | 8.21  | 6.64  | 5.08  | 5.43  | 7.58  | 8.49  | 9.63  | 11.70 |
|                                                                        | HDD50       | 4311   | 1050  | 892   | 692   | 262   | 32    | 0     | 0     | 0     | 7     | 134   | 445   | 797   |
|                                                                        | HDD65       | 7963   | 1515  | 1312  | 1153  | 688   | 293   | 97    | 14    | 39    | 186   | 515   | 889   | 1262  |
|                                                                        | CDD50       | 2226   | 0     | 0     | 5     | 26    | 223   | 420   | 612   | 539   | 308   | 86    | 7     | 0     |
|                                                                        | CDD65       | 401    | 0     | 0     | 0     | 1     | 20    | 67    | 161   | 113   | 37    | 2     | 0     | 0     |
|                                                                        | CDH74       | 2993   | 0     | 0     | 1     | 15    | 212   | 552   | 1215  | 712   | 275   | 11    | 0     | 0     |
|                                                                        | CDH80       | 630    | 0     | 0     | 0     | 1     | 40    | 123   | 294   | 131   | 41    | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 4.6    | 5.5   | 5.2   | 5.4   | 5.7   | 4.6   | 4.0   | 3.4   | 3.3   | 3.4   | 4.4   | 5.1   | 4.9   |
| (15) Precipitation                                                     | PrecAvg     | 47.6   | 3.4   | 2.8   | 3.2   | 3.6   | 4.0   | 4.3   | 4.9   | 4.6   | 4.3   | 4.4   | 3.7   | 4.2   |
|                                                                        | PrecMax     | 57.6   | 6.5   | 6.8   | 4.9   | 7.0   | 8.3   | 7.6   | 6.9   | 8.4   | 8.1   | 7.8   | 5.3   | 8.4   |
|                                                                        | PrecMin     | 39.0   | 1.5   | 1.5   | 1.4   | 0.7   | 2.0   | 2.0   | 2.1   | 1.9   | 1.7   | 2.0   | 1.6   | 2.0   |
|                                                                        | PrecStd     | 4.5    | 1.0   | 1.0   | 0.7   | 1.6   | 1.5   | 1.5   | 1.1   | 1.3   | 1.5   | 1.8   | 0.9   | 1.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 53.3  | 50.0  | 67.2  | 76.2  | 84.7  | 87.6  | 90.0  | 87.2  | 84.8  | 75.7  | 64.8  | 53.8  |
|                                                                        |             | MCWB   | 50.9  | 45.2  | 58.1  | 59.0  | 65.4  | 71.4  | 75.2  | 73.9  | 72.7  | 64.4  | 58.0  | 49.2  |
|                                                                        | 2%          | DB     | 46.3  | 42.8  | 53.9  | 68.0  | 79.7  | 83.6  | 85.9  | 83.5  | 80.3  | 70.7  | 59.3  | 47.4  |
|                                                                        |             | MCWB   | 43.1  | 38.6  | 46.0  | 52.9  | 64.3  | 69.7  | 71.4  | 70.7  | 68.7  | 61.6  | 53.7  | 44.3  |
|                                                                        | 5%          | DB     | 40.7  | 38.9  | 48.4  | 62.0  | 75.4  | 80.1  | 83.3  | 80.7  | 77.4  | 66.5  | 54.0  | 43.1  |
|                                                                        |             | MCWB   | 37.8  | 35.7  | 42.6  | 51.4  | 61.9  | 66.8  | 71.0  | 69.1  | 67.6  | 59.2  | 49.2  | 41.0  |
|                                                                        | 10%         | DB     | 36.3  | 35.8  | 43.9  | 56.9  | 71.1  | 76.7  | 80.5  | 78.0  | 73.5  | 61.9  | 49.1  | 39.2  |
|                                                                        |             | MCWB   | 34.1  | 33.6  | 39.6  | 47.1  | 59.6  | 65.1  | 69.4  | 68.5  | 65.8  | 55.7  | 45.2  | 37.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 51.7  | 47.2  | 58.6  | 61.2  | 71.1  | 74.6  | 77.8  | 76.5  | 74.4  | 66.9  | 59.3  | 50.2  |
|                                                                        |             | MCDB   | 53.1  | 49.6  | 66.4  | 72.5  | 79.3  | 84.0  | 86.9  | 83.6  | 82.1  | 72.1  | 63.9  | 52.3  |
|                                                                        | 2%          | WB     | 43.7  | 39.1  | 47.6  | 56.7  | 67.5  | 72.2  | 74.7  | 73.1  | 71.9  | 64.0  | 54.9  | 44.8  |
|                                                                        |             | MCDB   | 45.4  | 42.1  | 51.5  | 64.8  | 75.8  | 80.0  | 81.7  | 80.4  | 78.4  | 69.8  | 59.5  | 46.7  |
|                                                                        | 5%          | WB     | 37.9  | 35.9  | 43.1  | 52.1  | 64.5  | 70.0  | 72.9  | 71.3  | 69.4  | 60.2  | 50.1  | 41.2  |
|                                                                        |             | MCDB   | 39.9  | 38.2  | 46.8  | 60.0  | 71.4  | 76.6  | 79.9  | 77.5  | 75.1  | 65.0  | 53.8  | 43.0  |
|                                                                        | 10%         | WB     | 34.2  | 33.4  | 39.6  | 48.6  | 61.5  | 67.6  | 71.2  | 69.9  | 67.0  | 56.8  | 46.1  | 37.3  |
|                                                                        |             | MCDB   | 36.2  | 35.7  | 44.0  | 55.5  | 68.7  | 74.1  | 77.7  | 75.9  | 72.7  | 61.2  | 49.4  | 39.0  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 16.9  | 17.3  | 17.0  | 18.0  | 19.8  | 18.9  | 18.8  | 18.0  | 19.4  | 16.2  | 14.5  | 13.8  |
|                                                                        |             | MCDBR  | 18.9  | 19.2  | 22.4  | 28.1  | 25.6  | 23.2  | 21.4  | 20.7  | 22.5  | 22.5  | 21.2  | 15.9  |
|                                                                        |             | MCWBR  | 17.2  | 16.0  | 15.5  | 17.2  | 13.6  | 12.0  | 10.3  | 10.4  | 12.1  | 13.5  | 16.2  | 14.5  |
|                                                                        |             | MCDBR  | 18.2  | 16.8  | 19.4  | 23.7  | 21.3  | 18.7  | 17.9  | 16.9  | 20.1  | 19.1  | 19.8  | 15.2  |
|                                                                        | 5% WB       | MCWBR  | 17.1  | 14.6  | 14.8  | 16.3  | 13.6  | 10.6  | 9.7   | 9.3   | 12.2  | 13.1  | 16.8  | 14.5  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.259 | 0.291 | 0.324 | 0.358 | 0.380 | 0.426 | 0.432 | 0.412 | 0.374 | 0.332 | 0.308 | 0.289 |
|                                                                        | taud        |        | 2.377 | 2.268 | 2.327 | 2.364 | 2.359 | 2.230 | 2.220 | 2.272 | 2.371 | 2.494 | 2.482 | 2.351 |
|                                                                        | Ebn at Noon |        | 274   | 282   | 288   | 286   | 282   | 268   | 264   | 265   | 267   | 266   | 253   | 246   |
|                                                                        | Edh at Noon |        | 23    | 31    | 35    | 36    | 38    | 43    | 43    | 40    | 33    | 25    | 21    | 22    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 457   | 755   | 1141  | 1376  | 1616  | 1777  | 1788  | 1586  | 1259  | 727   | 442   | 318   |
|                                                                        | RadStd      |        | 52    | 79    | 91    | 163   | 155   | 137   | 132   | 74    | 103   | 66    | 57    | 27    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (46 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +106  | N/A   |

Nomenclature: See separate page